# Homework 2

STATISTICAL MODELING AND LEARNING

due: **Friday, January 30, 2026 at 10:00a**

Instructions: Please submit solutions on Populi as a knitted markdown pdf. For handwritten work, please scan your solution pages and include in the final pdf. Homeworks can be completed individually or in groups of up to three people. Names of all group members must be included on the writeups.

**Problem 1: ISL Exercises:**

Chapter 3 Exercises – Problems 3, 4, 14

Chapter 7 Exercises – Problem 9

**Problem 2:** This problem explores the conceptual understanding of interaction terms in regression models. Suppose we are studying factors that predict Instagram follower count.

a. Consider a regression model predicting Instagram follower count from number of posts per week and whether the account is verified. Let $Y$ = follower count, $X_{\text{posts}}$ = number of posts per week, and $X_{\text{verified}}$ = indicator for verified account (where $X_{\text{verified}} = 1$ if verified, 0 if not verified). Suppose we fit two models:

   - Model 1 (no interaction): Follower Count $= \beta_0 + \beta_1 X_{\text{posts}} + \beta_2 X_{\text{verified}} + \varepsilon$
   - Model 2 (with interaction): Follower Count $= \beta_0 + \beta_1 X_{\text{posts}} + \beta_2 X_{\text{verified}} + \beta_3 (X_{\text{posts}} \times X_{\text{verified}}) + \varepsilon$

   Interpret the coefficient $\beta_3$ in Model 2. What does it mean if $\beta_3$ is positive? What does it mean if $\beta_3$ is zero?

b. In Model 2 from part (a), write down the predicted follower count for an account posting 5 times per week in two cases: (i) verified ($X_{\text{verified}} = 1$), and (ii) not verified ($X_{\text{verified}} = 0$). Show how the difference between these predicted follower counts depends on both $\beta_2$ and $\beta_3$.

c. Consider a study examining the relationship between number of hashtags per post, account type, and follower count. Let $Y$ = follower count, $X_{\text{hashtags}}$ = number of hashtags per post, and $X_{\text{business}}$ = indicator for business account (where $X_{\text{business}} = 1$ for business account, 0 for personal account). A researcher fits a model with an interaction term:

$$\text{Follower Count} = \beta_0 + \beta_1 X_{\text{hashtags}} + \beta_2 X_{\text{business}} + \beta_3 (X_{\text{hashtags}} \times X_{\text{business}}) + \varepsilon$$

   - What does it mean if $\beta_3 > 0$ in the context of this problem?
   - What does it mean if $\beta_3 = 0$? How does this relate to Model 1 from part (a)?
   - If $\beta_1 = 200$, $\beta_2 = 5000$, and $\beta_3 = 150$, what is the predicted follower count for a personal account using 10 hashtags per post? What about a business account using 10 hashtags per post?

d. In multiple regression, an interaction between two variables means that the effect of one variable depends on the level of the other variable. Give an example (different from those above) of a real-world scenario where you would expect an interaction effect to be important. Explain why an interaction would make sense in this context.

**Problem 3:** This problem investigates variable transformations and demand modeling.

Download the file `beer-demand.csv` from the course website. This dataset contains weekly sales data for beer at a supermarket chain. The variables include:

- `PRICE12PK`, `PRICE18PK`, `PRICE30PK`: Prices (in dollars) for 12-packs, 18-packs, and 30-packs, respectively
- `CASES12PK`, `CASES18PK`, `CASES30PK`: Number of cases sold for 12-packs, 18-packs, and 30-packs, respectively

a. Focus on the 18-pack data. Create a scatter plot of cases sold (`CASES18PK`) versus price (`PRICE18PK`). Does the relationship appear linear? If not, describe the nature of the relationship.

b. Fit a simple linear regression model predicting `CASES18PK` from `PRICE18PK`. Examine the residual plots. What assumptions appear to be violated?

c. Fit a quadratic model by including both `PRICE18PK` and $\texttt{PRICE18PK}^2$ as predictors. Compare this model to the linear model from part (b) using $R^2$ and residual standard error. Is the quadratic term significant? Does the quadratic model improve the fit?

d. Apply log transformations to both variables and fit a linear regression model. Report the fitted model equation and interpret the coefficient on price.

e. In economics, price elasticity of demand is defined as the percentage change in quantity demanded for a 1% increase in price.

   (i) Mathematically, price elasticity is defined as:

$$\eta = \frac{dQ}{dP} \times \frac{P}{Q}$$

   where $Q$ is quantity demanded and $P$ is price. Starting from the log-log model $\log(Q) = \beta_0 + \beta_1 \log(P) + \varepsilon$, show that $\beta_1 = \eta$. (Hint: Differentiate both sides with respect to $\log(P)$, and recall that $\frac{d\log(Q)}{d\log(P)} = \frac{dQ/dP}{Q/P} = \frac{dQ}{dP} \times \frac{P}{Q}$.)

   (ii) What is the interpretation of the slope coefficient $\beta_1$ in your log-log model from part (d)? Explain how this interpretation relates to the definition of price elasticity.

f. Compare the models from parts (b), (c), and (d). Which model would you prefer for prediction? Justify your answer using appropriate metrics (e.g., $R^2$, residual standard error, diagnostic plots).

g. (Optional exploration) How does demand for 18-packs relate to the prices of other pack sizes? Fit a model predicting `CASES18PK` from `PRICE18PK`, `PRICE12PK`, and `PRICE30PK`. Interpret the coefficients. Do these results make economic sense?